Dev Patel

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Summary

Data Analyst with an M.S. in Business Analytics specializing in demand forecasting, inventory optimization, and applied AI. Builds forecasting, simulation, and optimization models in Python, SQL, and R for supply chain and business decisions.

Education

California State University, East Bay **Aug 2024 - May 2026**
Master of Science - Business Analytics, GPA - 3.8/4.0 *Hayward, CA*

California State University, Stanislaus **Aug 2019 - May 2024**
Bachelor of Science - Mathematics, GPA - 3.6/4.0 *Turlock, CA*

Experience

Financial Analyst Intern **May 2024 - Aug 2024**
City of Modesto *Modesto, CA*

- Analyzed monthly budget-to-actual performance for **10** departments within a **$228.6M** General Fund using **SQL** and **Excel**, identifying **$1.3M** in unfavorable spending variances and preparing corrective-action summaries for managers.
- Reconciled **300+** accounts, catching **$420K** in misclassified or duplicate transactions before month-end close.
- Built an **Excel** variance dashboard for 5 departments, saving **12 hours** of monthly reporting and **2 days** of review.
- Built **linear regression** and **ARIMA** models in **Python** to forecast city expenditures for budget planning.

Admissions Data Analyst **Dec 2023 - Aug 2024**
California State University, Stanislaus *Turlock, CA*

- Analyzed application data from student populations worldwide using **SQL** and **Excel**.
- Led outreach campaigns using predictive analytics, driving a **10%** Spring 2024 enrollment increase above target.
- Developed interactive **Tableau** dashboards - providing key insights that supported recruitment efforts and secured an increased budget for international student marketing.

Projects

Agentic AI for Inventory Optimization **Jan 2026 - May 2026**

- Trained a LightGBM global forecasting model on **50M+** rows of Walmart M5 retail data, achieving holdout RMSE of **1.79** (R^2 **0.65**); its inventory policies cut simulated stockouts by **68%** and raised ROI by **156%** vs. baseline.
- Engineered a **500-path** Monte Carlo simulation and a Mixed-Integer Linear Programming optimizer (Pyomo/HiGHS) to set cost-optimal (s, Q) policies under budget and service-level constraints, cutting portfolio operating cost by **25%**.
- Integrated a context-grounded Claude Sonnet AI assistant into a multi-page Streamlit app, enabling managers to query optimization results and run real-time demand-shock simulations in plain English with no data-science expertise.
- Technologies: Python, pandas, scikit-learn, LightGBM, Pyomo, SciPy, NumPy, Streamlit, Plotly, Anthropic Claude API

Traffic Forecasting for City of San Francisco Using Time-Series Analysis **Dec 2024 - Feb 2025**

- Developed a traffic forecasting system for the city of San Francisco using time-series models (ARIMA, ETS, SNaive, Prophet), achieving **90%** accuracy in traffic flow predictions over a 24-hour forecasting period.
- Trained and validated machine learning models in **R** and **Python**, using fable and prophet packages, with **RMSE** of **4.9** and **MAPE** reduction of **18%**.
- Processed **66,000+** SF traffic records, optimizing preprocessing and visualization for traffic management.
- Technologies: R, fable, Prophet, Python, pandas, ARIMA/ETS

Breast Cancer Diagnosis Using Supervised & Unsupervised Learning **Aug 2024 - Dec 2024**

- Built an L1-penalized SVM classifier in Python (scikit-learn) with 5-fold cross-validation, achieving a **93%** F1 score (**94%** precision, **92%** recall) and outperforming K-means and spectral clustering baselines (**88%** F1).

Technical Skills

Languages & Tools: Python, SQL, R, MATLAB, Mathematica, Java, JavaScript, Git, Jupyter
Machine Learning & Forecasting: scikit-learn, LightGBM, TensorFlow, PyTorch, Linear Regression, ARIMA, Prophet
Optimization & AI: Pyomo, HiGHS, SciPy, MILP, Monte Carlo Simulation, Anthropic Claude API
Data Analysis & Visualization: pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit, Tableau, Power BI, Excel
Databases & Data Engineering: Microsoft SQL Server, MySQL, PostgreSQL, Azure Data Factory